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BOOTCAMP DATA_

< RUSH 4 - USER SEGMENTATION />



BOOTCAMP DATA

Introduction

Social networks are under pressure to prove they are not harming their users. Regulators ask questions, the press publishes studies, and advertisers do not want to be associated with a platform that makes people miserable. Instagram's Innovation Department wants its own answer, based on its own data, before someone else provides one.

Your role

Context

Instagram's Innovation Department has hired your team to study the link between how people use the platform and their well-being. The department has two goals. First, measure whether and how usage relates to indicators such as stress, sleep and happiness. Second, understand the different kinds of users on the platform, so that product teams can design features for each of them.

The well-being and health information comes from an opt-in research panel: users who agreed to take part in a wellness study, and whose self-reported answers were linked to their account. The rest of each record — activity on the app and profile information — is held for every Instagram user. The department will present your conclusions to Instagram's leadership. Anything you claim will be challenged.

Objective

Deliver a segmentation of Instagram users that is grounded in their usage, explains how usage relates to well-being, and is usable by product teams. Defend it in front of a jury.

Logistics

Item	Detail
Group	5 people
Duration	5 days
Tools	Python or R, in a notebook
Data	Two extracts of 200,000 users each

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Expectations

The data

External consultants never see real users. Instagram's privacy team gave you a synthetic replica of the panel: artificial users generated to reproduce the statistical structure the team chose to preserve. Your recommendations, however, will be applied to real users.

The department sent two versions of the replica: one of them went through a cleaning pass that the privacy team did not document. They do not fully agree with each other. Which one you trust, on which fields, is your decision, and it must be documented.

Framing

Nobody has told you what a good segmentation is. Defining the criteria by which your segmentation should be judged, and showing that it meets them, is part of what is evaluated. At least some of those criteria must be checkable on the data.

You may build your segments by predicting a well-being indicator from usage, by clustering users into profiles, or both. Whichever route you take, the result must be segments, and you must explain why your route answers the department's question better than the alternative. Only panel members have well-being data, but product teams will want to place every user in a segment: your segments must be assignable to any user from the data Instagram holds on every user, and the notebook must show how a new user would be assigned.

What the segmentation must add

The department already knows that some users spend far more time on the app than others. Your segmentation must show what it tells them that ranking users by time spent does not — or demonstrate, with evidence, that nothing more can be said. The same goes for well-being: an indicator that turns out to have no link with usage is a finding.

The raw extracts describe what users do, one column at a time. The variables you build from them must be justified by what they add on the data, not by the story they tell.



Conclusions must state what the data does not allow you to claim, including anything about cause and effect, anything a panel of volunteers cannot tell you about all users, and anything a synthetic replica cannot tell you about real ones.



Some methods do not scale to 200,000 users.

Data protection

Panel members agreed to a wellness study. Product teams now want to use your segments to target features at users, which is not the same thing. For users in the European Union and the United Kingdom, including users under 18, your deliverables must take a position on what is legally possible and under which conditions: which variables your segments may be built on if they are to be used for targeting, and whether placing a user in a segment can by itself reveal something about their health.

The privacy team cannot guarantee that the generator never reproduced a real user. The delivered notebook must not display any individual user's record, nor any result that isolates a handful of users.

Defence

Your team presents its work in **15 minutes** to a jury playing Instagram's leadership, followed by **10 minutes** of questions. The jury will ask questions you did not prepare for.



Every member of the team must be able to answer questions on any part of the work. A defence split into five independent monologues does not show a team result.

Deliverables

Deliverable	Content
Notebook	Data choices, exploration, feature engineering, segmentation, evaluation against your criteria, limits, data protection
Defence slides	Support for the 15-minute presentation
Oral defence	15 minutes of presentation, 10 minutes of questions

The notebook must run from top to bottom on the provided extracts and produce every result shown in the defence, identically on each run.

The group repository is part of the deliverables: its history must show how the work was shared, and its README must state who was responsible for what.

Resources

The two extracts are available in the student resources of the module (`rush-4/`).

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